In [1]:

```
import pandas as pd
```

In [2]:

```
dataset = pd.read_csv("Placement.csv")
```

In [3]:

```
dataset
```

Out[3]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	No	55.0
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	Yes	86.5
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	No	75.0
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	No	66.0
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	No	96.8
...	...	...	...	...	...	...	...	...	...	...	...
210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt	No	91.0
211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech	No	74.0
212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt	Yes	59.0
213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt	No	70.0
214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt	No	89.0

215 rows × 15 columns

In [4]:

```
dataset.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 215 entries, 0 to 214
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   sl_no           215 non-null    int64
 1   gender          215 non-null    str
 2   ssc_p           215 non-null    float64
 3   ssc_b           215 non-null    str
 4   hsc_p           215 non-null    float64
 5   hsc_b           215 non-null    str
 6   hsc_s           215 non-null    str
 7   degree_p        215 non-null    float64
 8   degree_t        215 non-null    str
 9   workex          215 non-null    str
 10  etest_p         215 non-null    float64
 11  specialisation  215 non-null    str
 12  mba_p           215 non-null    float64
 13  status          215 non-null    str
 14  salary          148 non-null    float64
```

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```
dtypes: float64(6), int64(1), str(8)
memory usage: 35.0 KB
```

In [5]:

```
dataset.isnull().sum()
```

Out[5]:

```
sl_no              0
gender             0
ssc_p              0
ssc_b              0
hsc_p              0
hsc_b              0
hsc_s              0
degree_p           0
degree_t           0
workex             0
etest_p            0
specialisation     0
mba_p              0
status             0
salary            67
dtype: int64
```

In [6]:

```
dataset.isna().sum()
```

Out[6]:

```
sl_no              0
gender             0
ssc_p              0
ssc_b              0
hsc_p              0
hsc_b              0
hsc_s              0
degree_p           0
degree_t           0
workex             0
etest_p            0
specialisation     0
mba_p              0
status             0
salary            67
dtype: int64
```

In [7]:

```
dataset.columns
```

Out[7]:

```
Index(['sl_no', 'gender', 'ssc_p', 'ssc_b', 'hsc_p', 'hsc_b', 'hsc_s',
       'degree_p', 'degree_t', 'workex', 'etest_p', 'specialisation', 'mba_p',
       'status', 'salary'],
      dtype='str')
```

In [8]:

```
dataset["gender"].dtypes
```

Out[8]:

```
<StringDtype(na_value=nan)>
```

In [9]:

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```
for columnName in dataset.columns:
    print(columnName)
```

```
sl_no
gender
ssc_p
ssc_b
hsc_p
hsc_b
hsc_s
degree_p
degree_t
workex
etest_p
specialisation
mba_p
status
salary
```

In [11]:

```
if (dataset["gender"].dtype=="O"):
    print("Qual")
else:
    print("Quan")
```

```
Quan
```

In [12]:

```
Quan = []
Qual = []
for columnName in dataset.columns:
    print(columnName)
    if (dataset[columnName].dtype=="O"):
        print("Qual")
    else:
        print("Quan")
```

```
sl_no
Quan
gender
Quan
ssc_p
Quan
ssc_b
Quan
hsc_p
Quan
hsc_b
Quan
hsc_s
Quan
degree_p
Quan
degree_t
Quan
workex
Quan
etest_p
Quan
```

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```
specialisation
Quan
mba_p
Quan
status
Quan
salary
Quan
```

In [13]:

```
Quan = []
Qual = []
for columnName in dataset.columns:
    print(columnName)
    if (dataset[columnName].dtype=="O"):
        print("Qual")
        Qual.append(columnName)
    else:
        print("Quan")
        Quan.append(columnName)
```

```
sl_no
Quan
gender
Quan
ssc_p
Quan
ssc_b
Quan
hsc_p
Quan
hsc_b
Quan
hsc_s
Quan
degree_p
Quan
degree_t
Quan
workex
Quan
etest_p
Quan
specialisation
Quan
mba_p
Quan
status
Quan
salary
Quan
```

In [14]:

```
Quan
```

Out[14]:

```
['sl_no',
 'gender',
 'ssc_p',
 'ssc_b',
```

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```
 'hsc_p',
 'hsc_b',
 'hsc_s',
 'degree_p',
 'degree_t',
 'workex',
 'etest_p',
 'specialisation',
 'mba_p',
 'status',
 'salary']
```

In [15]:

```
Qual
```

Out[15]:

```
[]
```

In [16]:

```
def quanqual(dataset):
    Quan = []
    Qual = []
    for columnName in dataset.columns:
        print(columnName)
        if (dataset[columnName].dtype=="O"):
            print("Qual")
            Qual.append(columnName)
        else:
            print("Quan")
            Quan.append(columnName)
            return Quan, Qual
```

In [17]:

```
Quan, Qual = quanqual(dataset)
```

```
sl_no
Quan
```

In [18]:

```
Quan
```

Out[18]:

```
['sl_no']
```

In [19]:

```
Qual
```

Out[19]:

```
[]
```

In [20]:

```
dataset
```

Out[20]:

	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p
0	1	M	67.00	Others	91.00	Others	Commerce	58.00	Sci&Tech	No	55.0
1	2	M	79.33	Central	78.33	Others	Science	77.48	Sci&Tech	Yes	86.5
2	3	M	65.00	Central	68.00	Central	Arts	64.00	Comm&Mgmt	No	75.0

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	sl_no	gender	ssc_p	ssc_b	hsc_p	hsc_b	hsc_s	degree_p	degree_t	workex	etest_p
3	4	M	56.00	Central	52.00	Central	Science	52.00	Sci&Tech	No	66.(
4	5	M	85.80	Central	73.60	Central	Commerce	73.30	Comm&Mgmt	No	96.8
...	...	...	...	...	...	...	...	...	...	...	...
210	211	M	80.60	Others	82.00	Others	Commerce	77.60	Comm&Mgmt	No	91.(
211	212	M	58.00	Others	60.00	Others	Science	72.00	Sci&Tech	No	74.(
212	213	M	67.00	Others	67.00	Others	Commerce	73.00	Comm&Mgmt	Yes	59.(
213	214	F	74.00	Others	66.00	Others	Commerce	58.00	Comm&Mgmt	No	70.(
214	215	M	62.00	Central	58.00	Others	Science	53.00	Comm&Mgmt	No	89.(

215 rows × 15 columns

In [21]:
```
dataset["hsc_p"].mean()
```

Out[21]:
```
np.float64(66.33316279069768)
```

In [22]:
```
dataset["hsc_p"].median()
```

Out[22]:
```
np.float64(65.0)
```

In [23]:
```
dataset["hsc_p"].mode()[0]
```

Out[23]:
```
np.float64(63.0)
```

In [24]:
```
dir(dataset)
```

Out[24]:
```
['T',
 '_AXIS_LEN',
 '_AXIS_ORDERS',
 '_AXIS_TO_AXIS_NUMBER',
 '_HANDLED_TYPES',
 '__abs__',
 '__add__',
 '__and__',
 '__annotations__',
 '__array__',
 '__array_priority__',
 '__array_ufunc__',
 '__arrow_c_stream__',
 '__bool__',
 '__class__',
 '__contains__',
 '__copy__',
 '__dataframe__',
 '__deepcopy__',
```

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```
'__delattr__',
'__delitem__',
'__dict__',
'__dir__',
'__divmod__',
'__doc__',
'__eq__',
'__finalize__',
'__firstlineno__',
'__floordiv__',
'__format__',
'__ge__',
'__getattr__',
'__getattribute__',
'__getitem__',
'__getstate__',
'__gt__',
'__hash__',
'__iadd__',
'__iand__',
'__ifloordiv__',
'__imod__',
'__imul__',
'__init__',
'__init_subclass__',
'__invert__',
'__ior__',
'__ipow__',
'__isub__',
'__iter__',
'__itruediv__',
'__ixor__',
'__le__',
'__len__',
'__lt__',
'__matmul__',
'__mod__',
'__module__',
'__mul__',
'__ne__',
'__neg__',
'__new__',
'__or__',
'__pandas_priority__',
'__pos__',
'__pow__',
'__radd__',
'__rand__',
'__rdivmod__',
'__reduce__',
'__reduce_ex__',
'__repr__',
'__rfloordiv__',
'__rmatmul__',
'__rmod__',
'__rmul__',
'__ror__',
'__round__',
'__rpow__',
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
'__rsub__',
'__rtruediv__',
'__rxor__',
'__setattr__',
'__setitem__',
'__setstate__',
'__sizeof__',
'__static_attributes__',
'__str__',
'__sub__',
'__subclasshook__',
'__truediv__',
'__weakref__',
'__xor__',
'_accessors',
'_accum_func',
'_agg_examples_doc',
'_agg_see_also_doc',
'_align_for_op',
'_align_frame',
'_align_series',
'_append_internal',
'_arith_method',
'_arith_method_with_reindex',
'_attrs',
'_box_col_values',
'_can_fast_transpose',
'_check_copy_deprecation',
'_check_inplace_and_allows_duplicate_labels',
'_check_label_or_level_ambiguity',
'_clip_with_one_bound',
'_clip_with_scalar',
'_cmp_method',
'_combine_frame',
'_consolidate',
'_consolidate_inplace',
'_construct_axes_dict',
'_construct_result',
'_constructor',
'_constructor_from_mgr',
'_constructor_sliced',
'_constructor_sliced_from_mgr',
'_dir_additions',
'_dir_deletions',
'_dispatch_frame_op',
'_drop_axis',
'_drop_labels_or_levels',
'_ensure_valid_index',
'_find_valid_index',
'_flags',
'_flex_arith_method',
'_flex_cmp_method',
'_from_arrays',
'_from_mgr',
'_get_agg_axis',
'_get_axis',
'_get_axis_name',
'_get_axis_number',
'_get_axis_resolvers',
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
'_get_block_manager_axis',
'_get_bool_data',
'_get_cleaned_column_resolvers',
'_get_column_array',
'_get_index_resolvers',
'_get_item',
'_get_label_or_level_values',
'_get_numeric_data',
'_get_value',
'_get_values_for_csv',
'_getitem_bool_array',
'_getitem_multilevel',
'_getitem_slice',
'_gotitem',
'_hidden_attrs',
'_indexed_same',
'_info_axis',
'_info_axis_name',
'_info_axis_number',
'_info_repr',
'_init_mgr',
'_inplace_method',
'_internal_names',
'_internal_names_set',
'_is_homogeneous_type',
'_is_label_or_level_reference',
'_is_label_reference',
'_is_level_reference',
'_is_mixed_type',
'_iset_item',
'_iset_item_mgr',
'_iset_not_inplace',
'_iter_column_arrays',
'_ixs',
'_logical_func',
'_logical_method',
'_maybe_align_series_as_frame',
'_metadata',
'_mgr',
'_min_count_stat_function',
'_module_source',
'_needs_reindex_multi',
'_pad_or_backfill',
'_reduce',
'_reduce_axis1',
'_reindex_axes',
'_reindex_multi',
'_reindex_with_indexers',
'_rename',
'_replace_columnwise',
'_repr_data_resource_',
'_repr_fits_horizontal_',
'_repr_fits_vertical_',
'_repr_html_',
'_repr_latex_',
'_reset_cache',
'_sanitize_column',
'_select_dtypes_indices',
'_series',
```

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```
'_set_axis',
'_set_axis_name',
'_set_axis_nocheck',
'_set_item',
'_set_item_frame_value',
'_set_item_mgr',
'_set_value',
'_setitem_array',
'_setitem_frame',
'_setitem_slice',
'_shift_with_freq',
'_should_reindex_frame_op',
'_slice',
'_stat_function',
'_stat_function_ddof',
'_to_dict_of_blocks',
'_to_latex_via_styler',
'_typ',
'_update_inplace',
'_validate_dtype',
'_values',
'_where',
'abs',
'add',
'add_prefix',
'add_suffix',
'agg',
'aggregate',
'align',
'all',
'any',
'apply',
'asfreq',
'asof',
'assign',
'astype',
'at',
'at_time',
'attrs',
'axes',
'between_time',
'bfill',
'boxplot',
'clip',
'columns',
'combine',
'combine_first',
'compare',
'convert_dtypes',
'copy',
'corr',
'corrwith',
'count',
'cov',
'cummax',
'cummin',
'cumprod',
'cumsum',
'degree_p',
```

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```
'degree_t',
'describe',
'diff',
'div',
'divide',
'dot',
'drop',
'drop_duplicates',
'droplevel',
'dropna',
'dtypes',
'duplicated',
'empty',
'eq',
'equals',
'etest_p',
'eval',
'ewm',
'expanding',
'explode',
'ffill',
'fillna',
'filter',
'first_valid_index',
'flags',
'floordiv',
'from_arrow',
'from_dict',
'from_records',
'ge',
'gender',
'get',
'groupby',
'gt',
'head',
'hist',
'hsc_b',
'hsc_p',
'hsc_s',
'iat',
'idxmax',
'idxmin',
'iloc',
'index',
'infer_objects',
'info',
'insert',
'interpolate',
'isetitem',
'isin',
'isna',
'isnull',
'items',
'iterrows',
'itertuples',
'join',
'keys',
'kurt',
'kurtosis',
```

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```
'last_valid_index',
'le',
'loc',
'lt',
'map',
'mask',
'max',
'mba_p',
'mean',
'median',
'melt',
'memory_usage',
'merge',
'min',
'mod',
'mode',
'mul',
'multiply',
'ndim',
'ne',
'nlargest',
'notna',
'notnull',
'nsmallest',
'nunique',
'pct_change',
'pipe',
'pivot',
'pivot_table',
'plot',
'pop',
'pow',
'prod',
'product',
'quantile',
'query',
'radd',
'rank',
'rdiv',
'reindex',
'reindex_like',
'rename',
'rename_axis',
'reorder_levels',
'replace',
'resample',
'reset_index',
'rfloordiv',
'rmod',
'rmul',
'rolling',
'round',
'rpow',
'rsub',
'rtruediv',
'salary',
'sample',
'select_dtypes',
'sem',
```

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```
'set_axis',
'set_flags',
'set_index',
'shape',
'shift',
'size',
'skew',
'sl_no',
'sort_index',
'sort_values',
'specialisation',
'squeeze',
'ssc_b',
'ssc_p',
'stack',
'status',
'std',
'style',
'sub',
'subtract',
'sum',
'swaplevel',
'tail',
'take',
'to_clipboard',
'to_csv',
'to_dict',
'to_excel',
'to_feather',
'to_hdf',
'to_html',
'to_iceberg',
'to_json',
'to_latex',
'to_markdown',
'to_numpy',
'to_orc',
'to_parquet',
'to_period',
'to_pickle',
'to_records',
'to_sql',
'to_stata',
'to_string',
'to_timestamp',
'to_xarray',
'to_xml',
'transform',
'transpose',
'truediv',
'truncate',
'tz_convert',
'tz_localize',
'unstack',
'update',
'value_counts',
'values',
'var',
'where',
```

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```
 'workex',
 'xs']
```

In [ ]:

```

```

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